

Day 06 Hybrid Notes

Triangle constraints and coordinate triangle problems | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets

- I can use triangle inequality to decide whether side lengths form a triangle.
- I can connect larger sides with larger opposite angles.
- I can use the triangle midsegment theorem.

Part 1: Cloze Notes - Essential Ideas

1. The sum of any two side lengths in a triangle must be greater than the _____.
2. The largest angle is opposite the largest _____.
3. The smallest side is opposite the smallest _____.
4. A midsegment connects the midpoints of two sides of a _____.
5. A triangle midsegment is parallel to the third side and half its _____.
6. A median connects a vertex to the midpoint of the opposite _____.
7. Coordinate triangle problems often use distance and _____.

Part 2: Outline Notes - Big Picture

- A. Triangle inequalities tell whether a triangle is _____.
- B. Side-angle relationships help rank parts without solving _____.
- C. Midsegments create parallel lines and proportional _____.
- D. Coordinate formulas help verify triangle _____.

Connection to Prior Math

1. What algebra skill shows up today?

2. What diagram or graph skill shows up today?

3. What is one place this could appear outside math class?

Most Important Reminder

Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

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Guided setup + active learning

Part 3: Guided Setup Practice

1. Can 5, 7, and 13 form a triangle?
Setup/work: _____
Answer: _____
2. A midsegment is parallel to a side of length 18. Find the midsegment length.
Setup/work: _____
Answer: _____

Part 4: Error Analysis

A student says 3, 4, and 7 form a triangle because $3 + 4 = 7$.
What went wrong?

Correct idea:

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Before You Solve Checklist

- ☐ I labeled the diagram or graph.
- ☐ I chose the correct relationship or formula.
- ☐ I substituted values carefully.
- ☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that _____.

Sentence 2: One thing I need to remember when solving is _____.

Part 5C: Application Problem

A triangular sign has sides 12 in, 16 in, and 20 in. Which angle is largest?
Setup: _____
Answer: _____
Sentence answer: _____